Soluções - Folha 3

1a) $D_f = \mathbb{R}\setminus\{0\}$; $D_g = \{x \in \mathbb{R} : x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}\}$

b) $D_{f\circ g} = D_g$;

c) $(f\circ g)\left(\frac{2\pi}{3}\right) = \frac{3}{4}$

2a) $D_f = \{x \in \mathbb{R} : x \neq \frac{k\pi}{2}, k \in \mathbb{Z}\}$

b) Não tem zeros

c)

3) $D_f =]-\pi, \pi[$

4a) $\frac{\pi}{6}$ b) $\frac{\pi}{6}$ c) $\frac{\pi}{4}$ d) $\frac{\pi}{3}$ e) $-\frac{\pi}{3}$

f) $-\frac{\sqrt{3}}{2}$ g) $\frac{3\pi}{6}$ h) $-\frac{\sqrt{3}}{3}$

5a) $D_f = \mathbb{R}$ b) $D_f = \mathbb{R}$

c) $D_f = \{x \in \mathbb{R} : x \neq \pi + 2k\pi, k \in \mathbb{Z}\}$

d) $D_f = \{x \in \mathbb{R} : x \neq k\pi, k \in \mathbb{Z}\}$

e) $D_f = [0, 1]$ f) $D_f = [-1, 0]$

g) $D_f = \mathbb{R}$ h) $D_f = \mathbb{R}$

i) $D_f = [-2, -1]$ j) $D_f = [-1, -\frac{1}{2}] \setminus \{\frac{\pi}{4} - \frac{1}{2}\}$

6 a) $D_f = [-1,0]$; $D'_f = [0,\pi]$; $x = -1$

b) $D_f = \mathbb{R}$; $D'_f =]-\pi/3, \frac{2\pi}{3}[$; $x = -\frac{\sqrt{3}}{9}$

c) $D_f = \mathbb{R} \setminus \{-1\}$; $D'_f =]-\pi/2, \pi/2[\setminus \{0\}$; Não tem zeros

d) $D_f = [1-\sqrt{3}, 1+\sqrt{3}]$; $D'_f = [-\pi/2, \pi/6]$; $x=0 \vee x=1$

7 a) $D_f = [-2,2]$; $D'_f = [\frac{\sqrt{2}}{2}-\pi, \frac{\sqrt{2}}{2}+\pi]$

$D_g = \mathbb{R}$; $D'_g = [-1,7]$

b) $g^{-1}(x) = -\pi/3 + \arcsen\left(\frac{3-x}{4}\right)$

8 a) $D_f = [-2/3, 2/3]$; $D'_f = [\frac{1}{2}, \frac{1}{2}-2\pi]$

$D_g = \mathbb{R}$; $D'_g =]-\pi/4, 3\pi/4[$

b) $f^{-1}: [1/2, 1/2-2\pi] \longrightarrow [-2/3, 2/3]$

$x \mapsto \frac{2}{3}\cos\left(\frac{1-2x}{4}\right)$

$g^{-1}:]-\pi/2, 3\pi/4[\longrightarrow \mathbb{R}$

$x \mapsto -3 + \cotg\left(\frac{4x+\pi}{8}\right)$

9a) $D_f = [-1/2, 1/2]$; $D'_f = [\frac{\sqrt{2}+3\pi}{2}, \frac{\sqrt{2}-3\pi}{2}]$

b) $f^{-1}: [\frac{\sqrt{2}+3\pi}{2}, \frac{\sqrt{2}-3\pi}{2}] \longrightarrow [-1/2, 1/2]$

$x \mapsto \frac{1}{2}\sen\left(\frac{\sqrt{2}-2x}{6}\right)$

c) $x = \pi/4$

d) $x = \frac{1}{2}\sen\left(\frac{-\pi+2\sqrt{2}}{6}\right)$